MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 7ABP7A AND 7ABP19A

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic focus, magnetic deflection, aluminized screens

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb2	Ehk	Rg1	Eb1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	Vdc	ft
Maximum:	6.9	0, -200	770	11,000	±180	Note 1	1,100	10,000
Minimum:	5.7	---	---	7,000	---	---	-500	---
TEST CONDITIONS:	6.3	Adjust	300	7,000	---	---	Focus	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
1141	Pressure (implosion)	Both		---	---	---	---	---	---
5111	Vibration	Both		---	---	Width	---	2.0	mm
1331	Direct-interelectrode capacitance	Both	Cathode to all g1 to all g2 to all	--- --- ---	--- --- ---	Ck Cg1 Cg2	--- --- ---	8.0 10.0 10.0	pF pF pF
5201	Electrode current (grid No. 2)	Both	Ec1 = 0	---	---	Ic2	-15	+15	µAdc
5223	Modulation	P7A	Ib2 = 100 µAdc; overscanned defocused pattern	---	---	ΔEc1	---	32	Vdc
		P19A	Ib2 = 2 µAdc; overscanned defocused pattern	---	---	ΔEc1	---	15	Vdc
(D) 1216	Base material insulating quality	Both	Zone 5	---	---	---	---	---	---
	Quality conformance inspection, part 1								
5201	Voltage breakdown	Both		---	---	---	---	---	---
5201	Voltage breakdown (electrostatic types)	Both		---	---	---	---	---	---
5206	Gas ratio	Both	See note 4	---	---	Gr	---	0.25	---
5101	Neck straightness	Both	See note 3	---	---	---	---	---	---
5106	Bulb, screen, and faceplate quality	Both		---	---	---	---	---	---
5231	Spot position (magnetic deflection)	Both		---	---	---	---	9.52	mm
5236	Zero-bias anode current (magnetic deflection)	P7A		---	---	---	---	---	---
5241	Grid cutoff voltage	Both		---	---	Ec1	-33	-77	Vdc
5251	Grid No. 1 leakage current	Both		---	---	---	---	---	---
---	Focusing electrode voltage	P7A	Ib2 = 100 µAdc (see note 2)	---	---	Eb1	0	250	Vdc
		P19A	Ib2 = 2 µAdc (see note 2)	---	---	Eb1	0	300	Vdc
	Quality conformance inspection, part 2								
1301	Heater current	Both		---	---	If	540	660	mA
5101	Side terminal and base alignment	Both	Pin position No. 3	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
5101	Neck and bulb alinement (magnetic types)	Both		---	---	---	---	---	---
Ⓓ 1101	Secureness of base, cap, or insert	Both		---	---	---	---	---	---
5101	Face tilt	Both		---	---	---	---	---	---
5216	Cathode illumination	Both		---	---	---	---	---	---
5216	Stray light emission (conventional types)	Both	Eb2 = 11,000 Vdc; Ec2 = 770 Vdc; Eb1 = 0	---	---	---	---	---	---
5221	Screens	Both		---	---	---	---	---	---
5226	Line width A (electro-static deflection)	P7A	Ib2 = 100 μAdc	---	---	---	---	---	---
		P19A	Ib2 = 2 μAdc	---	---	---	---	0.50	mm
5251	Heater-cathode leakage current	Both		---	---	---	---	0.35	mm
5251	Grid No. 2 leakage current	Both		---	---	---	---	---	---
Ⓓ 1111	Base pin solder depth	Both		---	---	---	---	---	---
1105	Permanence of marking	Both		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	Both	Group C; t = 500 hours minimum; Eb2 = 11,000 Vdc; Ec2 = 770 Vdc; Eb1 at focus	---	---	---	---	---	---
		P7A	Ib2 = 60 μAdc	---	---	---	---	---	---
		P19A	Ib2 = 2 μAdc	---	---	---	---	---	---
---	Life-test end points	P7A	Line width A	---	---	---	---	---	---
			Modulation	---	---	Width ΔEc1	---	0.50	mm
		P19A	Line width A	---	---		---	32	Vdc
			Modulation	---	---	Width ΔEc1	---	0.35	mm
		Both	Grid No. 1 leakage current	---	---		---	15	Vdc
			Heater-cathode leakage current	---	---	---	---	---	---
			Grid No. 2 leakage current	---	---	---	---	---	---
			Stray light emission (conventional types)	---	---	---	---	---	---

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NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg; when Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 Meg.
2. Use an overscanned 35 to 105 line raster pattern with not less than 1/4-inch (6.35 mm) spacing between adjacent lines; adjust for best focus at center of tube.
3. Cylinder shall be 5-inches (127.0 mm) long and 1.503-inches (139.78 mm) maximum inside diameter.
4. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

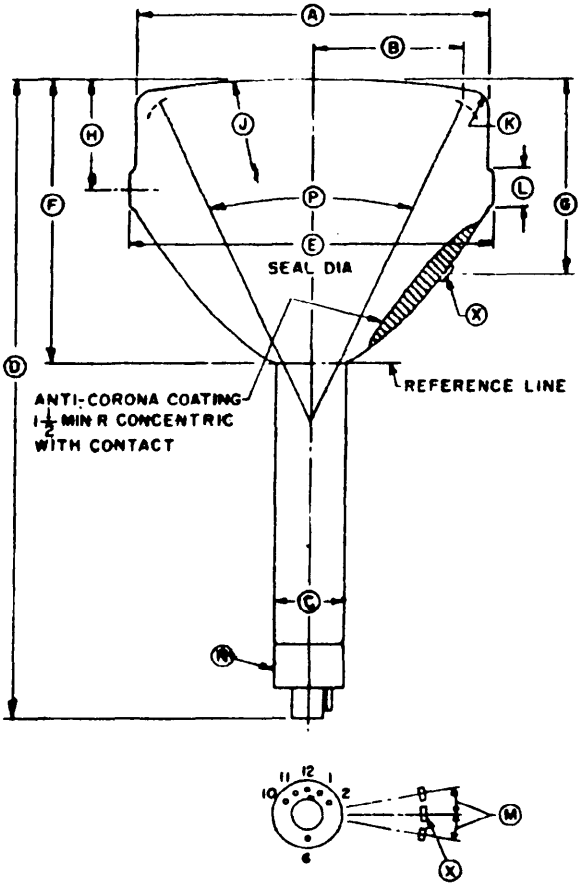
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

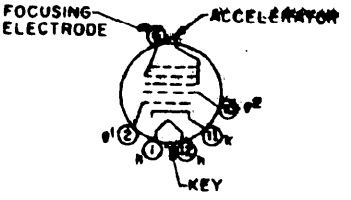
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2529)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
N	Base: B6-63	
X	Contact Cap J1-21	
Quality conformance inspection, part 2		
A	7.062(179.37)	7.312(185.72)
B	See note a	
C	1.375 (34.93)	1.500 (38.10)
D	13.000(330.20)	13.500(342.90)
E		7.312(185.72)
F	5.750(146.05)	6.125(155.58)
G	3.938(100.03)	4.438(112.73)
M		10°
Reference dimensions (see note b)		
J	24.00(609.60)R	
K	.375(9.53) R	
H	2.219(56.36)	
P	50°	
L	.750 (19.05)	

NOTES:
a. The minimum useful screen radius shall be not less than 3 inches(76.2 mm).
b. Reference dimensions are for information only and are not required for inspection purposes.
c. The bulb shall be a J57-1/2C or J57-1/2D type.
d. Do not handle tube by the part of the bulb having the anticorona coating.
e. Dimension C applies to the entire length of the neck below the reference line. The reference line is determined by the point where a gauge 1.500 (38.10 mm) ±.003(.08 mm) inches inside diameter and 2 inches(50.8 mm) in length will stop against the bulb body.

FIGURE 1. Outline drawing for electron tube types 7ABP7A and 7ABP19A.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-88806

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be of corrosion-resistant metal or treated to resist corrosion.